Poster presentation at Focus Meeting 14
General Assembly of the International Astronomical Union
Vienna, August 2018

The International Astronomical Union and the Arab World: Bridging Communities in Science and Outreach

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Over the last hundred years, the International Astronomical Union has played a crucial role in bridging communities and developing space science and outreach in the Arab world. Algeria was, as part of France, a founding member of the IAU in 1919. Egypt joined the Union in 1925, and other Arab countries followed suit in the second half of the 20th century. Lebanon first joined in 1954, Iraq in 1976, and Morocco and Saudi Arabia in 1988. Individual IAU members were also important, especially for countries such as Iraq, which at times were at the margins of the international scientific community. In 2010, no Iraqi institution, but seven individual astronomers were still members of the IAU. Especially active was Athem Alsabti, who had first joined the union in 1976. After leaving Iraq during the late 1990s, he became part of the IAU's Program Group for the World-Wide Development of Astronomy. In honour of his work, the IAU named the asteroid 10478 Alsabti after him in 2001. The collaboration between IAU members on different continents has been essential for outreach and the training of astronomers in the Arab world. Zdeněk Kopal at the University of Manchester attracted many students from Egypt, Iraq, and Libya between 1950s and 1980s. Kopal and his colleagues initiated the International Schools for Young Astronomers, which have been supported by the IAU. In 1967, the University of Manchester hosted the first of these schools with twelve students, including two from Egypt. Subsequently, Egypt and Morocco each hosted two International Schools for Young Astronomers between 1981 and 2004. While military conflicts and light pollution have formed significant challenges, increasing numbers of science museums, booming amateur clubs and arid mountains in the region provide opportunities for future outreach. Collaboration with local science fiction societies could further contribute to fascination for astronomy among people with limited access to planetaria or telescopes.

Growing professional associations engaged in outreach: **Arab Union for Astronomy and Space Sciences** in Jordan (established 1998), **Arab Astronomical Society** in Morocco (established 2016).

Booming amateur societies: **Sirius Astronomy Association** in Algeria (established 1996), **Jeddah Astronomical Society** in Saudi Arabia (established 2007), **Oman Astronomical Society** (established 2008).

Potential partners for reaching out to people without easy access to planetaria and telescopes: the **Egyptian Society for Science Fiction** (established 2012) and the Syrian magazine ***Science Fiction*** (published since 2008).

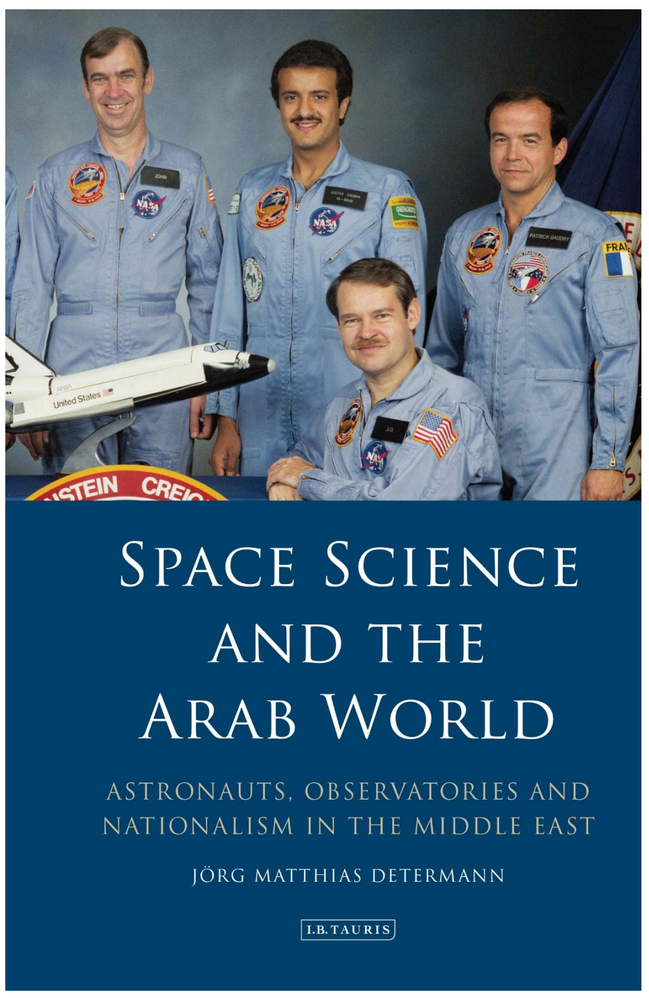


Opportunities for future outreach through the **Sharjah Center for Astronomy & Space Sciences** in the United Arab Emirates (opened 2015), **Fakieh Planetarium** in Saudi Arabia (opened 2016), **Katara Planetarium** in Qatar (scheduled to open end of 2018).





This poster is based in part on my book *Space Science and the Arab World: Astronauts, Observatories and Nationalism in the Middle East*.



Activities of the **Egyptian Society for Science Fiction** have included publications, competitions, and salons. Key figures include the founder Hosam Elzembely and Emad El-Din Aysha. Elzembely is author of the novels *The Great Space Saga: The Half-Humans*, *The Wondrous Planet: First Dialogue with an Intelligent Virus*, and *America 2030* (first published in 2001).





I am currently writing a book with the working title *Islam and Extraterrestrial Life: Astrobiology and Science Fiction in the Muslim World.* I thank Fajr Aamir, Haldun Faruk Gümüş, İrfan Batur, Mosammat Samiha Sadeka, and Walli Ullah for their contributions and assistance.